

AnyCam 3.5.0-rc1.0 Audit Report

Release audit - 04 October 2026 - DVR channel count, one camera database - tag 3.5.0-rc1.0

Scope: phase 3.5.0 of the build plan: D1 (CrystalHeeler, 2026-10-04: automatic, the channel limit from what the DVR reports) and D2 (write the plan, add tests, then merge). Built in the untested run on top of 3.4.0-rc1.0. This build is not field-tested.

1. D1: the DVR's channel count

Part	Design
When	After the password is accepted on a Lorex/Dahua DVR, before the channel walk.
Ask	GET /cgi-bin/devVideoInput.cgi?action=getCollect (answer result=N) and /cgi-bin/magicBox.cgi?action=getProductDefinition&name;=MaxRemoteInputChannels (answer table.MaxRemoteInputChannels=N), on the DVR's HTTP address. Basic login, then Digest on a 401, with the password just accepted.
Use	The larger of the two answers from 1 to 256 sets the channels walked; the count is saved on the card as dvr_channels.
No answer	16 channels, as before, with a log line that says so.

Before 3.5.0 every DVR was walked for channels 1 to 16: 7 needless probes on the 8-channel Lorex DVR on test system B, and no cards past 16 on a larger NVR. The lockout counter of this brand counts failed logins only; the password used here has just been accepted.

2. D2: one camera database

The plan is docs/Camera_DB_Merge_Plan.md, written first. In short:

- Each CAMERA_DB brand holds its former STREAM_DB entries as "streams" (38 brands, 40 entries; Hikvision also holds ezviz, Dahua also holds imou).
- STREAM_DB is built from them at start-up, in the old order ("rank"), so a tie between two equally long keywords goes the same way.
- The data was moved by a script that compared the built table with the old one: SHA-256 of its JSON e14a73de... before and after.
- The scan's port list is built from the database plus the documented extra ports; it is the same 54 ports today.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 16 files; build inputs checked online; the CAMERA_DB throttle field check.
Python integration	PASS	542 of 542; 10 new (AF1 to AF6), including a fake DVR that asks for Digest login. Z7 now says the DVR does not report a count, and still walks 2 to 16.
JavaScript behaviour	PASS	158 of 158, unchanged.

Suite	Result	Scope
Undefined names	PASS	16 files, 0.
Not run	NOTED	No field test (CrystalHeeler's order). The image was not built. The Lorex DVR's real answer to the two requests is not known; the parsing follows Dahua's HTTP API format.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.4.0-rc1.0 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	16 files.
1.16 Refactoring	Behaviour kept	PASS	STREAM_DB byte for byte; port list equal; Z and Y tests unchanged.
1.17 Standard library	Data	PASS	No new dependency; one source for each fact.
1.18 Testing	Tests first	PASS	Plan, then tests, then the merge, as decided.
6.1 Security	DVR answer	PASS	The count is bounded (1 to 256); the password goes only to the DVR.
Rule 9 (privacy)	New text	PASS	Tests use made-up addresses only.

Known issues carried forward

- **D1:** no sample yet of what a known-empty channel sends; such a channel can get a card.
- **B6:** the Microseven waits until CrystalHeeler unlocks it.
- **C21:** the classic engine stays as the fallback.

Acceptance (field test)

- Test system B: after the password, the log says the DVR reports 8 channels, and only channels 2 to 8 are walked.
- The channel cards are the same as in 3.4.0-rc1.0.
- A scan finds the same cameras as before.

Files changed

File	Change
camera_db.py	Stream paths on the brands; the STREAM_DB builder.

File	Change
anycam_credentials.py	_dvr_channel_count, _dvr_parse_count; the walk uses the count.
anycam_scan.py	The port list built from the database.
docs/Camera_DB_Merge_Plan.md	New.
tests/	Section AF; Z7 with no reported count.
CHANGELOG.md, config.yaml, CLAUDE.md, docs	3.5.0-rc1.0; build plan.